NewsBot Group Reflection Journal

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Team: Sergio Sanchez & Richard Fox · **Course:** ITAI 2373 Natural Language Processing · **Project:** NewsBot Intelligence System (Mid-Term)

GitHub: Richard Fox – ITAI2373-NewsBot-Midterm · Sergio Sanchez – ITAI2373-NewsBot-Midterm

1. Team Collaboration

We worked as a two-person team on one shared notebook, passing versions back and forth and reviewing each other's changes. That review step caught the most problems: Sergio fixed a bug in the first version where new articles got placeholder zeros, and Rich caught a dataset that didn't fit the assignment and a report showing the wrong words per category.

Next time, we would agree on the dataset and do a full fresh test run together early in the week, instead of finding issues close to the deadline.

2. Technical Integration Challenges

Each step worked on its own; the hard part was making them work together and making sure the results were right, not just error-free.

- **Mixing features:** Naive Bayes can't take negative numbers, so we rescaled sentiment and length scores to 0–1 before combining them with TF-IDF. The final NewsBot class first fed new articles placeholder zeros; we fixed it to calculate real scores.
- **Data problems:** we removed 50 duplicate articles so none could land in both training and testing, and went back to the BBC dataset after a trial with forum posts (20 Newsgroups).
- **A misleading report:** the SVM's "top words" showed sale/offer/sell for most categories, because it stores weights per *pair* of categories. We used Logistic Regression's weights instead.
- **Lowercase text:** it helps classification but hides names from the tagger, so proper nouns scored near 0% and "us" was counted as a country 568 times. We could only work around this.

Speed was the other challenge. Running spaCy on 1,440 full-length articles was slow, so we switched to batch processing and turned off the parts of the model we didn't need for entity extraction. Getting the notebook to run cleanly in both Google Colab and a local Windows setup also took real time (package versions, NLTK data files, and a Kaggle download that kept failing), but it means anyone can now run it from top to bottom.

3. Business Value

NewsBot turns unread articles into sorted, searchable facts in about a second each. Who could use it:

- **Newsrooms and content teams:** tag and route articles automatically, sending only unclear ones to an editor.

- **Business intelligence:** track coverage of specific companies, people or countries, and whether it is positive or negative.
- **Risk and compliance:** spot negative news about partners, vendors or markets early.
- **Any text, not just news:** the same steps can sort customer emails, support tickets and survey comments.

The Gradio web app makes this usable by people who don't write code: paste an article, click Analyze, and see the category, tone and highlighted names.

Its most useful feature is knowing when to ask for help: all five new test articles were classified correctly, and the two it was unsure about (AI chips at 47%, a US Senate story at 45%) were flagged for human review. Sentiment needs care, though: 47% of Politics articles scored negative because politics covers disputes, not because the reporting is biased.

4. Individual Contributions

- **Richard Fox:** built the first end-to-end version on BBC data; moved the project back to BBC and removed the duplicates; fixed the top-features report and install issues; replaced pre-written findings with calculated results; wrote the analysis sections, summary and README.
- **Sergio Sanchez:** expanded every module's analysis and charts; combined features and compared three models with cross-validation; sped up entity extraction; rebuilt the NewsBot class with real features and confidence routing; wrote the test articles; built the Gradio web app.

5. Future Enhancements

A model only knows its training data: ours learned British politics from 2004–2005 ("labour", "tory", "blair"), so it was unsure about US and AI stories. What we would build next:

1. **Better data:** recent news from several countries, with original capitalization and real headlines. This fixes both the confidence problem and the name-finding problem.
2. **A transformer model (like BERT):** it reads words in context, so it should handle new topics better than word counts.
3. **Cleaner entities:** filter out number words like "first" and "two", and skip the headline so it isn't tagged as a company name.
4. **Trends over time:** store results in a database to track how coverage of a company or topic changes week to week.
5. **Web app upgrades:** upload a batch of articles and download the results as a spreadsheet.

6. Professional Development Impact

The code ran early; most of our effort went into making sure the results could be trusted. Three lessons we will take into future work:

- **Data quality comes first.** Duplicates, the wrong dataset and lowercase text mattered more than which model we picked.
- **"It runs" is not "it's right."** Two of our worst bugs caused no errors at all.
- **Explaining limits is part of the job.** Knowing why the model struggles with US news is as useful to a business as its 95.1% accuracy.

We will both use NewsBot as a portfolio project: raw text in, a working web app out, with honest notes on what it can and can't do.